MIL-T-62102E
9 March 1987
SUPERSEDING
MIL-T-62102D
16 December 1983

MILITARY SPECIFICATION

TRUCK TRACTORS: SEMITRAILER SPOTTER, WITH HALF CAB, DIESEL AND GASOLINE ENGINE DRIVEN, 32,000 GVW, 55,000 GCW, 4x2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers half cab, diesel and gasoline engine driven, four-wheel, two-rear-wheel drive commercial truck tractors used for spotting semitrailers. The truck tractor has a minimum gross vehicle weight (GVW) of 32,000 pounds under normal operating conditions. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these do uments shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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STANDARDS

FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

*	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.
	MS51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
	MS75020	- Connector, Plug, Electrical - 12-Contact, Intervehicular, 28-Volt, Waterproof.
	MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

SAE, INC.

SAE Standards and Recommended Practices

J245	- Engine Rating Code - Spark Ignition.
J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J688	- Truck Ability Prediction Procedure (DoD adopted).
* J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted).
J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
J816	- Engine Test Code - Spark Ignition and Diesel.
J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

* (Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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THE TIRE AND RIM ASSOCIATION, INC. (T&RA)
Yearbook.

(Application for copies of T&RA publications should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.10, the vehicles, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

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3.1.1.3 Drain plug. The drain plug for the rear axle differential shall be of the permanent magnet type.

3.1.1.4 Heavy duty cooling system. A heavy duty cooling system shall be furnished which will maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.5 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than two-quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the dash instrument panel. On tilt cab models, a radiator servicing access door shall be provided to allow verification of coolant level.

3.1.1.6 Towing devices. Not less than two hooks, loops, pins, or a center mounted eye for towing the vehicle shall be furnished on the front of the vehicle and not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the rear of the vehicle. The towing devices shall be frame mounted. Front towing devices shall have a total capacity of not less than 60,000 pounds and rear towing devices shall have a total capacity of not less than 35,000 pounds.

* 3.1.1.7 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51118. The pintle shall be installed on the chassis frame with reinforcements to transfer a tongue load of not less than 4,000 pounds and a drawbar load of not less than 40,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be not more than 4 inches forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 40,000 pounds and shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32-inch throat width) for a 5/8-inch chain. A lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle.

* 3.1.1.8 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

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3.1.1.9 Front fenders. Fenders shall be furnished over that portion of the front wheel assemblies not covered by the cab. The front, top, and the rear of the wheel assemblies shall be enclosed. Sufficient reinforcement shall be furnished to minimize vibration. Removable rubber splash shields shall be furnished at the rear of the fenders.

* 3.1.1.10 Wheel splash and stone throw protection. The vehicle shall have rigid half fenders over the rear wheels and removable rubber mud flaps to the rear of the rear wheels. The mud flaps and their extension supports shall be readily removable, to increase landing wheel clearance, without the use of hand tools. The half fenders need extend down only to the height of the centerline of the rear axle. A metal strip, not less than 1/8-inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or bolt nuts from damaging the mud flap. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; fifth wheel; and a full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds), and a payload to provide not less than the specified GVW.

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3.2.6 Gross combination weight. Gross combination weight (GCW) shall consist of truck tractor curb weight, operator (weight computed at 175 pounds), and the weight of a semitrailer, loaded to provide not less than the specified GCW. The fifth wheel shall be located so that with the truck tractor loaded to GVW, the load rating of the chassis components are not exceeded.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. Minimum ratings for the truck tractor shall be 32,000 GVW and 55,000 GCW.

* 3.2.8 Dimensions. Vehicle wheelbase shall be not more than 104 inches and shall provide a cab-to-axle dimension of not more than 84 inches. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 96 inches. The width measured over the tires shall be not more than 100 inches.

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with truck tractor loaded to specified GCW.

3.3.1.1 High speed gradeability. The vehicle shall ascend a continuous grade of 1.0 percent at 35 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

* 3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, using torque converter ratio and the highest numerical ratio in the transmission and shall provide a vehicle speed of not more than 4 mph.

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 45 mph (minus 1 percent). Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

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3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the tractor-semitrailer combination, loaded to specified GCW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349.

* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline. Net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J245 or SAE J1349.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to engine manufacturer's maximum recommended operating speed.

3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

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* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.7 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and for grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block, or lower coolant inlet hose. Engine thermostat with an operating range of 170° to 195°F shall be installed;

(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening;

(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperature as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F;

(d) For diesel engine driven vehicles, a fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

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3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system and an alternator with a rating of not less than 60 amperes shall be furnished. Engine starting system shall include a glow plug or an ether system. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. Unless otherwise specified, an alternator of not less than 40-ampere rated capacity and which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished. When a converter-type 24-volt system is provided (see 3.4.2.8), an alternator of not less than 60 amperes capacity which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20 and 393.22 through 393.26(d). The brake lights shall override the four-way emergency flasher.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. Signal units shall be visible and audible in combination with the towed vehicle and shall be visible when not in combination with the towed vehicle. Turn signal operating units shall have a visible and audible type flash indicator.

* 3.4.2.5 Semitrailer lighting cable. A semitrailer lighting cable conforming to SAE J1067 and incorporating a connector conforming to SAE J560 on the semitrailer end shall be provided. The cable shall be coiled and shall have an extended length of not less than 110 inches. The SAE J560 connector shall include a grip for withdrawing from the semitrailer receptacle. Stowage for the cable shall be provided by means of hook and hanging loop or protective holding bracket. When hook and loop method is used, the cable shall be furnished with a hanging loop, so attached as to insure that the plug is pointed down when the

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cable is stowed in the hook provided on the rear of the truck tractor cab or on a pogo stick type hose tender. The protective bracket, when furnished, shall be mounted at the rear of the cab, below the roof line, and shall hold the cable plug within the bracket housing to preclude entrance of weather within the terminals.

* 3.4.2.6 Trailer lighting cable. A trailer lighting cable, conforming to SAE J1067, shall be furnished for use with the lighting receptacle specified in 3.1.1.7. The cable shall be not less than 110 inches long and both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

* 3.4.2.7 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating at 80°F and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table II. Batteries shall be of the maintenance free type having the maintenance-free characteristics listed in W-B-131.

TABLE II. Batteries.

Engine	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1200
Gasoline	100	450

* 3.4.2.8 Auxiliary 24-volt system with trailer receptacle. When specified (see 6.2), an auxiliary 24-volt system with a trailer receptacle assembly shall be furnished. The auxiliary 24-volt system output capacity shall be not less than 12 amperes. Either a converter type (see 3.4.2.8.1) or an alternator type (see 3.4.2.8.2) system meeting specified requirements shall be furnished. A trailer receptacle conforming to MS75021-2, with cover assembly, shall be provided in an accessible location on the rear of the cab. A twelve-conductor truck tractor jumper cable not less than 10 feet long, with both ends of the cable equipped with connectors conforming to MS75020-1 and MS75020-2 shall be furnished. The cable assembly shall be stowed in the vehicle. The 24-volt, 12-ampere output service lighting circuit shall be connected through the appropriate lighting controls to terminals B, D, E, J and L of MS75021-2. A cab mounted hose tender shall be provided to accommodate and secure the 24-volt jumper cable.

* 3.4.2.8.1 Converter-type 24-volt system. The 12- to 24-volt converter(s) shall operate from the 12-volt battery (see 3.4.2.7).

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3.4.2.8.2 Alternator type 24-volt system. The alternator type 24-volt system shall be separate from the 12-volt vehicle lighting and ignition system and shall include:

(a) A nominal 24-volt alternator with not less than 25-ampere rated capacity and capable of providing not less than 7 amperes dc output at normal engine idle speed;
(b) Two 12-volt batteries with a combined capacity of at least 40 ampere-hours at a 20-hour rate or one 24-volt battery with at least 20 ampere-hours capacity at a 20-hour rate;
(c) A voltage regulating device;
(d) An ammeter, mounted on the instrument panel.

3.4.2.9 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tanks. Fuel tank(s) of not less than 50-gallon total capacity (43 gallons for California) shall be furnished. When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single gage.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust system, when stacked, shall be furnished with a heat shield and a hinged rain cap. On diesel engine driven vehicles the tail pipe shall be stacked on the curbside of the engine transversely in line with the rear half of the cab. All portions of the exhaust system above the body sheet metal shall be fully enclosed by a heat shield. On gasoline engine driven vehicles, all portions of the exhaust system which project to the rear of the cab shall be provided with a cover plate. The cover plate shall form a shield that will shed fluids, precluding spilled fuel from coming into contact with the exhaust system. A spark arrester shall be furnished, except on turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

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* 3.4.5 Automatic or semiautomatic transmission. An automatic or semiautomatic transmission shall be provided. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power takeoff opening(s). When specified (see 6.2), a hydraulic retarder shall be provided.

3.4.5.1 Power takeoff. A power takeoff shall be provided. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear reduction. The drive shaft shall be protected in a manner similar to that specified for buses by Federal Motor Carrier Safety Regulation 393.89.

3.4.7 Frame. Chassis frame shall be the manufacturer's standard for the vehicle furnished. Frame shall have sufficient structural strength and resisting bending moment to at least equal the loads imposed under the specified operating conditions with vehicle loaded to specified GVW. When frame reinforcements are required to meet the specified loads, the reinforcements shall extend at least from the rear of the front-suspension, rear hanger bracket to the front of the rear-spring, front hanger bracket. Crossmembers shall be provided at least rearward of the transmission and at the rear. The chassis frame rails shall be cut off immediately to the rear of the rear spring rear hanger bracket or the frame crossbar closest to the rear of this bracket.

3.4.8 Suspension. The vehicle shall be equipped with a front spring suspension system with components having a rated capacity at least equal to the loads imposed in each member, measured at the ground, with the vehicle loaded to specified GVW. When a spring suspension system is provided for the rear axle, the rear spring suspension system with components shall have a rated capacity at least equal to the load imposed on each member, measured at the ground with the vehicle loaded to the specified GVW. When the suspension system is rated at the spring pads, unsprung weight shall be deducted. Vehicle shall be equipped with hydraulic double-acting shock absorbers at the front wheels.

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3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. A single reduction rear axle shall be furnished.

3.4.9.1 Traction control. A traction control shall be furnished on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Tire, rim and disc wheel sizes shall be the same for all wheels on each vehicle. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

3.4.10.1 Tires. Unless otherwise specified (see 6.2), tires shall be steel belted radial ply tubeless type or tube type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. For tube tires, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

* 3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm);
(b) Air storage reservoir(s) with not less than 2,000 cubic inches total capacity, each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank;
(c) Foot control, suspended or treadle type;
(d) Air control valves;
(e) Air pressure gage, visible to the driver;

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(f) Low-air-pressure warning, visible and audible;
(g) Service brake stop lamp switch;
(h) Automatic moisture ejector;
* (i) Automatic slack adjusters.

* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have a capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Trailer brake control system. In addition to the components specified in 3.4.11.1 and 3.4.11.1.1, a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines;
(b) Coincident control of trailer brakes with prime mover foot control;
(c) Independent hand control for trailer brakes;
(d) Prime mover protection valve with dash control and automatic break-away feature;
(e) Trailer stoplight control operative with foot brake and with hand control for trailer brakes;
(f) Two SAE J844 coiled air brake hoses, not less than 110 inches in length when fully extended, equipped with coiled spring hose guards, and SAE J318 gladhand quick connectors on the trailer end of the hoses for towing semitrailers;
(g) Dummy gladhand connectors to retain these hoses when not in use;
(h) Two SAE J844 coiled air hoses, not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses for towing full trailers. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment;
* (i) Air connectors for full trailers with SAE J318 gladhand couplers at the rear of the truck tractor located to preclude interference with the full trailer. Air connectors shall be provided with a cover assembly and a security chain for each connector;
(j) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

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3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature except when spring applied emergency brake is furnished, additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. The manufacturer shall furnish a half width cab suited to spotter truck operations. Tilt type half cab, if furnished, shall include a tilting and locking mechanism. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. The half cab, conventional on tilt type, shall have a sedan type driver's side door and a sedan sliding or folding-leaf type rear door. Doors shall be equipped with locks operable from inside the cab. Driver's side door shall be equipped with an external key operated lock. Drip rails shall be installed above the cab doors, and above windows that open. The cab shall have an upholstered adjustable seat and back for the driver. The seat shall be capable of swiveling 90 degrees to the right, from normal driving position, to allow the driver easy entry and exit by the rear door. The color of the upholstery and the interior finish shall be compatible with the exterior color. White upholstery shall not be furnished. An interior dome light shall be provided. The window in the door on the driver's side shall be mechanically operated. The curb side window shall be either mechanically operated or shall be of the sliding type. The window in the rear door shall be capable of being opened. Means of opening will be dictated by the type of door supplied. Manufacturer's standard fastenings and one pair of seat belts shall be installed. Grab rails and treaded steps shall be provided on both sides of the vehicle to assist personnel in climbing into the cab and onto the truck-tractor deck plate. A suitable step and grab rail shall be provided at the front of the vehicle to facilitate cleaning of the windshield.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with a windshield wiper and a washer. The windshield wiper shall be of the multispeed type and operated by either air or an electric motor.

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* 3.4.15 Bumpers. A channel type heavy duty front bumper shall be provided.

3.4.16 Stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, anti-skid chains and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheel-nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator, to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Ash receptacle
(h) Sunvisor
(i) Hydraulic or pneumatic fifth wheel elevating and descending control

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(j) Fifth wheel lock and unlock mechanism located inside cab and accessible to seated driver
(k) Front door mounted armrest on driver's side
(l) Driver's compartment ventilator other than window
* (m) Tachometer
* (n) When specified (see 6.2), a sensor and load gage, measuring and indicating the load on the fifth wheel.

3.4.20 Rearview mirrors. The vehicle shall be equipped with outside rearview mirrors. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex area shall have not less than 20 square inches of reflective area. One mirror shall be mounted on the street side of the cab. The other shall be pole-mounted in a suitable position on the curbside of the vehicle to afford the driver proper rear vision when moving standard width trailers and semitrailers. Mirrors shall be friction and shockproof mounted and shall be fully adjustable. The streetside mirror shall have not less than three supporting arms.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be provided.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the instrument panel or in the engine compartment in a readable location.

3.5 Hydraulic lift fifth wheel. Unless otherwise specified (see 3.6), vehicle shall be equipped with a hydraulic lift fifth wheel. The hydraulic fifth-wheel unit shall be a power operated, all-steel, heavy duty item with a rated capacity to lift and support a load of not less than 50,000 pounds over the kingpin, and with a lifting range of not less than 14 inches from the lowest to the highest position. The fifth wheel and its mounting structure shall have a design capacity to withstand a drawbar pull of 60,000 pounds without residual deformation to the mounting, the fifth wheel or the truck-tractor structure. The fifth-wheel shall be of the fore and aft rocking type with a 36-inch diameter steel coupler plate supplied with beveled approach forks. The fifth-wheel unit shall conform to Federal Motor Carrier Safety Regulation 393.70(b).

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3.5.1 Fifth wheel subframe. The fifth wheel assembly shall be mounted to a subframe. The subframe (or tractor main frame) shall be provided with an inclined ramp presenting a continuous slope for semitrailer approach. The subframe shall be mounted to the vehicle chassis frame rails. Means shall be provided to preclude deformation of the subframe and the chassis frame rails. The subframe, fifth wheel assembly and its mounting shall be capable of accepting the loads imposed during coupling, uncoupling, ascending and descending ramps up to 14 degrees from horizontal and the effects of weight transfer up and down the specified grades.

3.5.2 Fifth wheel hydraulic system. The fifth wheel unit shall include a hydraulic system and mechanism arranged to actuate an elevating platform equipped with a fifth wheel; shall incorporate a pneumatic system to open a semi-automatic lock for an SAE J700 kingpin; shall embrace a remote control to permit all fifth wheel operations from within the vehicle cab; and shall include all components, such as pump, hose and fittings, to comprise a complete installation. The remote control shall be provided with a guard to preclude accidental actuation.

3.5.3 Clearance. The clearance from the centerline of the kingpin to the cab, or pogo stick type hose tender when furnished, with the fifth wheel at all elevated positions, shall be not less than 60 inches and not more than 72 inches. The landing wheel clearance from the centerline of the kingpin to a vertical plane on the outside edge of the rear tires, chassis frame rails and rear fenders, with the fifth wheel in the lowest position, shall be not more than 60 inches. Under no condition shall the kingpin be positioned to the rear of the centerline of the rear axle. The unladen level height from the ground to the top of the coupler plate of the installed fifth wheel unit with the unit in its lowest position shall be 48 inches, plus or minus 1 inch. Actuation through the full lifting range shall not cause the center of the kingpin lock to shift more than 6-1/4 inches, measured along the longitudinal centerline of the vehicle.

3.5.4 Frame stops. Unless otherwise specified, frame stops shall be provided. Frame stops shall be capable of holding the fifth wheel in an elevated position of 55 inches plus or minus 1 inch, without carrying the imposed load on the lifting cylinders. Stops shall be hinged and shall fold out. A caution plate shall be furnished and mounted in a conspicuous location in the driver's compartment. At least the following information shall be shown: WHEN TOWING SEMITRAILER IN AN ELEVATED POSITION FOR MORE THAN 1,000 FEET, FRAME STOPS ARE TO BE USED TO KEEP IMPOSED LOAD OFF THE LIFTING CYLINDERS. When specified (see 6.2), or at the manufacturer's option, the hydraulic fifth wheel lifting

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cylinder(s) shall be fitted with hydraulically locked, flow-check safety valves in lieu of frame stops. The safety valves shall preclude the fifth wheel being lowered except by hydraulic power and shall prevent shock-loading of the hydraulic system.

3.5.5 Deck plate. Except for vehicles furnished with saddle tanks, self-cleaning grating of sufficient structural strength for use by the operator in connecting air and electric lines between the tractor and a semitrailer shall be installed. The grating shall extend across and shall be bolted to the frame rails or structural members with sufficient overhang to cover and protect side mounted tanks and fittings. The grating shall be located as close to the cab as possible and shall extend rearward to the maximum extent practicable. Provisions to allow for access by personnel climbing onto the deck plate shall be furnished. Access through the grating for maintenance of fittings and other equipment shall be furnished.

* 3.5.6 Hose tender. When a tilt-type cab is furnished, a pogo stick type hose tender shall be provided behind the cab to accommodate and secure the semitrailer lighting cable and air hoses. When a conventional cab is furnished, a hose tender as specified in 3.4.2.5 may be provided in lieu of the pogo stick. The pogo stick type hose tender shall be mounted on the truck-tractor deck plate or chassis members.

3.6 Air operated lift fifth wheel. When specified (see 6.2), the vehicle shall be equipped with an air operated lift fifth wheel in lieu of a hydraulic lift fifth wheel. The air operated fifth wheel unit shall be a self-contained, all-steel, heavy duty item, with a capacity to lift and support a load of not less than 50,000 pounds over the kingpin, and with a lifting range of not less than 13 inches, from the lowest to the highest position. Actuation through the full lifting range shall not cause the center of the kingpin lock to shift more than 4 inches measured along the longitudinal centerline of the vehicle. The fifth wheel unit shall utilize a base furnished with an incline ramp presenting a continuous slope for a semitrailer approach, and provided with a means for "bolt-on" attachment onto the vehicle chassis frame. The fifth wheel and its mounting structure shall have a design capacity to withstand a drawbar pull of 60,000 pounds without residual deformation to the mounting, fifth wheel or the truck tractor structure. The fifth wheel installation shall conform to 3.5.1.

3.6.1 Air operated fifth wheel components. The air operated fifth wheel shall be furnished and installed with at least the following:

(a) Air reservoir tank(s), not less than 7.4 cubic feet total capacity, equipped with drain, safety and check valves between compressor and tank

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(b) Air compressor, not less than 12 cfm, air or water cooled, engine driven and engine lubricated, unloader-head type
(c) Automatic moisture ejection valve
(d) Two air starter valves to emit and expel air from the reservoir tank to the air bellows
(e) Three-way valve, for raising, lowering or holding the fifth wheel in any desired position. The valve control shall be mounted on the truck tractor instrument panel in an accessible location for the driver, when in a seated position. The control shall be provided with a guard to preclude accidental activation
(f) Snubbers to eliminate spring deflection
(g) Pneumatic system to open and lock the SAE J700 kingpin, operable from the truck tractor instrument panel.

3.6.2 Fifth wheel (air operated). The fifth wheel shall be of the fore and aft rocking type, with a 36-inch cast steel coupler plate with beveled approach forks. The fifth wheel assembly shall conform to Federal Motor Carrier Safety Regulations 393.70(b).

3.6.3 Clearances. The fifth wheel clearance measured from the centerline of the kingpin location to the cab, or pogo stick, with the fifth wheel at all elevated positions, shall be not less than 60 nor more than 69 inches. The landing wheel clearance to the vertical plane of the outside edge of the rear-most tire and the chassis frame rails, with the fifth wheel in its lowest position shall be not more than 64 inches. The unladen level height from ground level to the top of the fifth wheel plate shall be 52 inches plus or minus 1 inch.

3.6.4 Deck plate. A deck plate shall be provided in accordance with 3.5.5.

3.6.5 Hose tender. A hose tender shall be furnished conforming to 3.5.6.

* 3.7 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alinement of lights; adjustment of the engine, electrical and brake systems; filling and charging of battery; alinement of front wheels; inflation of all tires; cycling the fifth wheel assembly through its entire range of elevations; complete lubrication of chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a

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solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.8 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government

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representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and the distribution of curb weight on the front and rear axles. The imposed loading on front and rear axles shall be computed by the contractor and verified by the Government using the curb weight, the operator weight at 175 pounds, and the payload to provide the specified GVW. Calculated imposed loads on front and rear axles shall be utilized to evaluate the capability of the suspension, axles and tires furnished to meet performance requirements.

* 4.3.2 Road test. The first production vehicle shall be examined and road tested (less payload) by the contractor. The road test shall be for not less than 10 miles at speeds up to 45 mph.

4.3.3 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.4 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.7 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government for shuttling and spotting full trailers and semitrailers at depot and roll-on roll-off operations.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Concealed marking, if required (see 3.1.1.1).
(d) Rustproofing, if required (see 3.1.1.2).
(e) Tropical rustproofing, if required (see 3.1.1.2).
(f) Gasoline engine, if required (see 3.4.1.2).
(g) Silicone rubber hoses, if required (see 3.4.1.7).
(h) Power plant heaters and fuel warmer, if required (see 3.4.1.8).
(i) If a battery heater is not required (see 3.4.1.8).
(j) Auxiliary 24-volt system with trailer receptacle, if required (see 3.4.2.8).
(k) Hydraulic retarder, if required (see 3.4.5).
(l) Disc type wheels, if required (see 3.4.10).
(m) Other type tires, if required (see 3.4.10.1).
(n) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
(o) Tools, if required (see 3.4.16.1).
* (p) Load gage, if required (see 3.4.19(n)).
(q) Air horn, if required (see 3.4.21).
(r) Hydraulic fifth wheel safety check valve, if required (see 3.5.4).
(s) Air operated lift fifth wheel, if required (see 3.6).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirement for each vehicle model furnished under the contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheelbase, engine model number, and vehicle type. Computations should be based on vehicle operating under normal atmospheric pressure, in normal ambient air temperature, in still, dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Tables:

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Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00.
Table 3	- Rolling Factor. 1.003.
Table 4	- Area Factor. 0.225.
Table 5	- Velocity Factor. 85.7.
Table 6	- Altitude Factor. 1.00.
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 35 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 1.07.
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

* 6.4 Subject term (key word) listing.
Air operated fifth wheel
Half cab
Hydraulic fifth wheel
Semitrailer spotter
Shuttle tractor
Truck tractor.

6.5 Identification of changes. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Review activites:
Navy - MC
Air Force - 84

Preparing activity:
Army - AT

Project No. 2320-0294

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

DOCUMENT IDENTIFIER *(Number)* **AND TITLE**

MIL-T-62102E TRUCK TRACTORS: Semitrailer Spotter, With Half Cab, DED and GED, 32,000 GVW, 55,000 GCW, 4x2

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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